

Lesson Plan and PCPH Practice Problem Bank (Topic-Wise)

Course Information

Course: PCPH / Soft Skills

Program: B.Tech. Second Year (Artificial Intelligence & Machine Learning)

Semester: Odd Semester (2026–2027)

Course Structure

The course is divided into **single component** to ensure comprehensive learning and continuous skill development throughout the semester.

1. LeetCode Practice (PCPH) – 2 Hours per Week

Dedicated laboratory sessions will be conducted for solving **curated LeetCode programming problems**. The problem sets are organized topic-wise and aligned with the Data Structures and Algorithms to progressively enhance students' coding proficiency and logical thinking.

2. Programming Concepts and Problem Solving

This component focuses on mastering fundamental programming concepts through hands-on practice. Students will learn to analyze problems, design algorithms, implement optimized solutions, and evaluate their computational efficiency.

The major topics include:

- Searching Algorithms
 - Sorting Algorithms
 - Array
 - String
 - Linked List
 - Stack
 - Queue
 - Two Pointer Technique
 - Sliding Window
 - Additional Problem Statements
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Recommended Practice Platforms

Students are encouraged to regularly practice coding problems on the following platforms:

- **LeetCode**
- HackerRank
- CodeChef
- Codeforces

1. Searching Algorithm

Linear Search, Binary Search

2. Sorting Algorithm

Selection Sort, Insertion Sort, Bubble Sort.

3. Array

#	Problem Name	Difficulty	Direct LeetCode Link
1	Two Sum	Easy	leetcode.com/problems/two-sum
2	Best Time to Buy and Sell Stock	Easy	leetcode.com/problems/best-time-to-buy-and-sell-stock
3	Contains Duplicate	Easy	leetcode.com/problems/contains-duplicate
4	Maximum Subarray	Medium	leetcode.com/problems/maximum-subarray
5	Product of Array Except Self	Medium	leetcode.com/problems/product-of-array-except-self
6	Rotate Array	Medium	leetcode.com/problems/rotate-array
7	Merge Intervals	Medium	leetcode.com/problems/merge-intervals
8	Find All Numbers Disappeared in an Array	Easy	leetcode.com/problems/find-all-numbers-disappeared-in-an-array
9	Subarray Sum Equals K	Medium	leetcode.com/problems/subarray-sum-equals-k
10	Next Permutation	Medium	leetcode.com/problems/next-permutation

2. String

#	Problem Name	Difficulty	Direct LeetCode Link
1	Valid Anagram	Easy	leetcode.com/problems/valid-anagram
2	Valid Palindrome	Easy	leetcode.com/problems/valid-palindrome
3	Longest Common Prefix	Easy	leetcode.com/problems/longest-common-prefix
4	First Unique Character in a String	Easy	leetcode.com/problems/first-unique-character-in-a-string
5	Reverse Words in a String	Medium	leetcode.com/problems/reverse-words-in-a-string
6	Group Anagrams	Medium	leetcode.com/problems/group-anagrams
7	Longest Palindromic Substring	Medium	leetcode.com/problems/longest-palindromic-substring
8	String to Integer (atoi)	Medium	leetcode.com/problems/string-to-integer-atoi
9	Count and Say	Medium	leetcode.com/problems/count-and-say
10	Find the Index of the First Occurrence in a String	Easy	leetcode.com/problems/find-the-index-of-the-first-occurrence-in-a-string

3. Linked List

#	Problem Name	Difficulty	Direct LeetCode Link
1	Reverse Linked List	Easy	leetcode.com/problems/reverse-linked-list
2	Linked List Cycle	Easy	leetcode.com/problems/linked-list-cycle
3	Merge Two Sorted Lists	Easy	leetcode.com/problems/merge-two-sorted-lists
4	Remove N-th Node From End of List	Medium	leetcode.com/problems/remove-nth-node-from-end-of-list
5	Reorder List	Medium	leetcode.com/problems/reorder-list
6	Intersection of Two Linked Lists	Easy	leetcode.com/problems/intersection-of-two-linked-lists
7	Middle of the Linked List	Easy	leetcode.com/problems/middle-of-the-linked-list
8	Add Two Numbers	Medium	leetcode.com/problems/add-two-numbers
9	Linked List Cycle II	Medium	leetcode.com/problems/linked-list-cycle-ii
10	Palindrome Linked List	Easy	leetcode.com/problems/palindrome-linked-list

4. Stack

#	Problem Name	Difficulty	Direct LeetCode Link
1	Valid Parentheses	Easy	leetcode.com/problems/valid-parentheses
2	Implement Queue using Stacks	Easy	leetcode.com/problems/implement-queue-using-stacks
3	Min Stack	Medium	leetcode.com/problems/min-stack
4	Evaluate Reverse Polish Notation	Medium	leetcode.com/problems/evaluate-reverse-polish-notation
5	Daily Temperatures	Medium	leetcode.com/problems/daily-temperatures
6	Next Greater Element I	Easy	leetcode.com/problems/next-greater-element-i
7	Decode String	Medium	leetcode.com/problems/decode-string
8	Asteroid Collision	Medium	leetcode.com/problems/asteroid-collision
9	Online Stock Span	Medium	leetcode.com/problems/online-stock-span
10	Simplify Path	Medium	leetcode.com/problems/simplify-path

5. Queue

#	Problem Name	Difficulty	Direct LeetCode Link
1	Implement Stack using Queues	Easy	leetcode.com/problems/implement-stack-using-queues
2	Design Circular Queue	Medium	leetcode.com/problems/design-circular-queue
3	Number of Recent Calls	Easy	leetcode.com/problems/number-of-recent-calls
4	Moving Average from Data Stream	Easy	leetcode.com/problems/moving-average-from-data-stream
5	Design Circular Deque	Medium	leetcode.com/problems/design-circular-deque
6	Task Scheduler	Medium	leetcode.com/problems/task-scheduler
7	Dota2 Senate	Medium	leetcode.com/problems/dota2-senate
8	Reveal Cards In Increasing Order	Medium	leetcode.com/problems/reveal-cards-in-increasing-order
9	Time Needed to Buy Tickets	Easy	leetcode.com/problems/time-needed-to-buy-tickets
10	Find Winner of Circular Game	Medium	leetcode.com/problems/find-the-winner-of-the-circular-game

6. Two Pointer Approach

#	Problem Name	Difficulty	Direct LeetCode Link
1	Two Sum II - Input Array Is Sorted	Medium	leetcode.com/problems/two-sum-ii-input-array-is-sorted
2	3Sum	Medium	leetcode.com/problems/3sum
3	Container With Most Water	Medium	leetcode.com/problems/container-with-most-water
4	Remove Duplicates from Sorted Array	Easy	leetcode.com/problems/remove-duplicates-from-sorted-array
5	Move Zeroes	Easy	leetcode.com/problems/move-zeroes
6	Valid Palindrome II	Easy	leetcode.com/problems/valid-palindrome-ii
7	Squares of a Sorted Array	Easy	leetcode.com/problems/squares-of-a-sorted-array
8	Sort Colors	Medium	leetcode.com/problems/sort-colors
9	Is Subsequence	Easy	leetcode.com/problems/is-subsequence
10	4Sum	Medium	leetcode.com/problems/4sum

7. Sliding Window

#	Problem Name	Difficulty	Direct LeetCode Link
1	Maximum Average Subarray I	Easy	leetcode.com/problems/maximum-average-subarray-i
2	Longest Substring Without Repeating Characters	Medium	leetcode.com/problems/longest-substring-without-repeating-characters
3	Longest Repeating Character Replacement	Medium	leetcode.com/problems/longest-repeating-character-replacement
4	Permutation in String	Medium	leetcode.com/problems/permutation-in-string
5	Minimum Size Subarray Sum	Medium	leetcode.com/problems/minimum-size-subarray-sum
6	Find All Anagrams in a String	Medium	leetcode.com/problems/find-all-anagrams-in-a-string
7	Max Consecutive Ones III	Medium	leetcode.com/problems/max-consecutive-ones-iii
8	Fruit Into Baskets	Medium	leetcode.com/problems/fruit-into-baskets
9	Subarray Product Less Than K	Medium	leetcode.com/problems/subarray-product-less-than-k
10	Maximum Erasure Value	Medium	leetcode.com/problems/maximum-erasure-value

Additional Problem Statements

LeetCode Links (Weeks 1–13)

Below are the direct links to the additional LeetCode problem statements referenced directly in the core 13-week schedule:

- **Pascal's Triangle:** leetcode.com/problems/pascals-triangle
- **Reverse String:** leetcode.com/problems/reverse-string
- **Reverse Vowels of a String:** leetcode.com/problems/reverse-vowels-of-a-string
- **Climbing Stairs:** leetcode.com/problems/climbing-stairs
- **Sort an Array (Merge Sort practice):** leetcode.com/problems/sort-an-array
- **N-Queens:** leetcode.com/problems/n-queens
- **Reverse Linked List II:** leetcode.com/problems/reverse-linked-list-ii
- **Rotate List:** leetcode.com/problems/rotate-list
- **Sliding Window Maximum:** leetcode.com/problems/sliding-window-maximum
- **Symmetric Tree:** leetcode.com/problems/symmetric-tree
- **Maximum Depth of Binary Tree:** leetcode.com/problems/maximum-depth-of-binary-tree
- **Diameter of Binary Tree:** leetcode.com/problems/diameter-of-binary-tree
- **Validate Binary Search Tree:** leetcode.com/problems/validate-binary-search-tree
- **Convert Sorted Array to Binary Search Tree:** leetcode.com/problems/convert-sorted-array-to-binary-search-tree
- **Number of Islands:** leetcode.com/problems/number-of-islands
- **Clone Graph:** leetcode.com/problems/clone-graph
- **Network Delay Time:** leetcode.com/problems/network-delay-time
- **Min Cost to Connect All Points:** leetcode.com/problems/min-cost-to-connect-all-points